

Smart Attendance System Using Face Recognition

Team Details:

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Abstract:

The Smart Attendance System is an automated attendance management solution that leverages face detection and facial recognition techniques to identify and mark student attendance accurately and efficiently. Traditional attendance methods are time-consuming, prone to human error, and susceptible to manipulation. This project addresses these limitations by implementing an AI-based solution capable of recognizing students in real-time, even within large classroom settings. During the enrollment phase, clear and high-resolution facial images of students are captured to generate unique facial embeddings that numerically represent each individual's distinguishing facial features. These embeddings, along with student roll numbers and batch details, are securely stored in a database for future comparison. During the attendance phase, a single group image of the class is captured using a standard camera. The RetinaFace algorithm detects all faces present in the frame, and each detected face is cropped and processed by the custom face recognition module. This module generates embeddings from each cropped face and compares them with the stored embeddings using cosine similarity to identify students and automatically mark their attendance.

Objectives:

- To design and develop a fully automated attendance system based on facial recognition.
- To implement RetinaFace for highly accurate and efficient face detection under variable conditions such as lighting, occlusion, and distance.
- To create a custom feature extraction and embedding generation algorithm without relying on OpenCV.
- To maintain a structured face embedding database linked to student roll numbers for fast and reliable retrieval.
- To ensure robust recognition despite moderate changes in appearance (e.g., hairstyle, spectacles, expression).
- To optimize the system for real-time performance, accuracy, and scalability across large classrooms.

Proposed Methodology:

- Data Collection & Enrollment – Capture clear frontal facial images of each student. Generate and store facial embeddings along with roll numbers and batch information in the database.

- **Face Detection (RetinaFace)** – Use the RetinaFace algorithm for detecting all visible faces in an image. It utilizes multi-scale anchor-based detection and contextual feature aggregation to handle variations in lighting, angle, and occlusion.
- **Feature Extraction (Custom Code)** – Develop a custom feature extraction module that encodes facial characteristics into a fixed-length vector (embedding) using CNN or mathematical methods (HOG, LBP, PCA).
- **Face Recognition** – Compute similarity between new embeddings and stored embeddings using cosine similarity or Euclidean distance.
- **Attendance Marking** – Store the attendance record (roll number, date, and time) in a database or file system.
- **Optimization** – Apply resolution checks, super-resolution models, and adaptive similarity thresholds for enhanced accuracy.

Tech Stack:

Component	Technology / Framework
Programming Language	Python
Face Detection	RetinaFace
Feature Extraction & Embedding Generation	Custom-built CNN / Mathematical approach (HOG, LBP, PCA)
Similarity Measurement	Cosine Similarity / Euclidean Distance
Database	SQLite / MySQL
Backend Framework	Flask / FastAPI
Image Processing	NumPy, Pillow
Model Training Dataset	LFW, VGGFace2

Algorithms Used:

- **RetinaFace** – For accurate and real-time face detection with multi-scale anchor-based feature extraction.
- **Custom Feature Extraction Algorithm** – CNN or mathematical model for deriving distinctive face embeddings.
- **Cosine Similarity / Euclidean Distance** – For comparing embeddings and deciding recognition matches.
- **Super-Resolution Algorithm (Optional)** – To restore low-quality facial crops before recognition.

Expected Outcomes:

- A complete, real-time Smart Attendance System capable of recognizing multiple students from a single group photo.
- A reliable and explainable feature extraction and recognition pipeline developed independently (without OpenCV).
- High recognition accuracy across different lighting conditions and appearance changes.
- Substantial reduction in human effort and attendance marking time.
- A modular, scalable, and privacy-preserving solution for educational institutions.
- Comprehensive technical documentation including architecture, algorithms, and database schema.